

ASSIGNMENTS

Class-3

1. Define pivot and free variables with example.
2. Identify pivot and free variables for the following systems.
 - i) $x + 3y - 6z = -10$, $2x + y - 5z = -8$, $6x - 3y + 3z = 0$
 - ii) $2x + y + 3z = 1$, $2x + 6y + 8z = 3$, $6x + 8y + 18z = 5$
3. Choose the number c so that (if possible) the ranks are (i) 1 (ii) 2 (iii) 3

i) $A = \begin{bmatrix} 1 & 4 & 2 \\ 1 & 2 & c \\ c & 1 & 2 \end{bmatrix}$ ii) $B = \begin{bmatrix} 1 & 2 & 2 \\ 1 & c & 1 \end{bmatrix}$